



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Review of Thermochemistry (Ch.5) And Thermodynamics (Ch. 19)

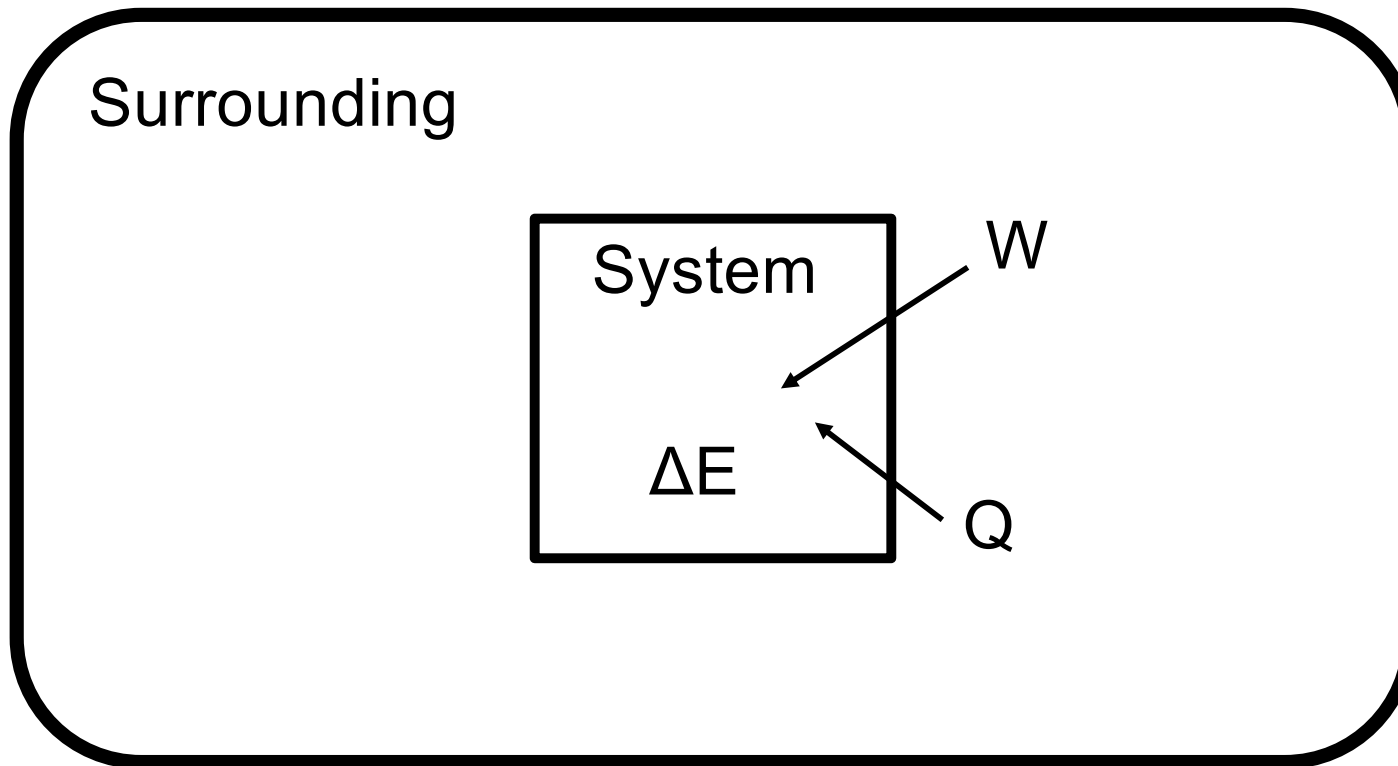
Instructor: Prof. Daigo Miyajima
E-mail: dmiyajima@cuhk.edu.cn
Office: RX1101

TA: Chenyu Yang
Xinying Jing
Dong Zhang
Jiaxiang Wang
Haoyu Feng

Nov. 11th, 2025

Heat and Work

First Law of Thermodynamics



No focus on E_p, E_k

$$\Delta E = Q + W$$

$$\Delta E = \cancel{\Delta k} + \cancel{\Delta V} + \Delta U$$

U: Internal Energy Thermochemistry

Momentum energy of molecules
Push



Heat and Work

$$\Delta U = Q + W$$

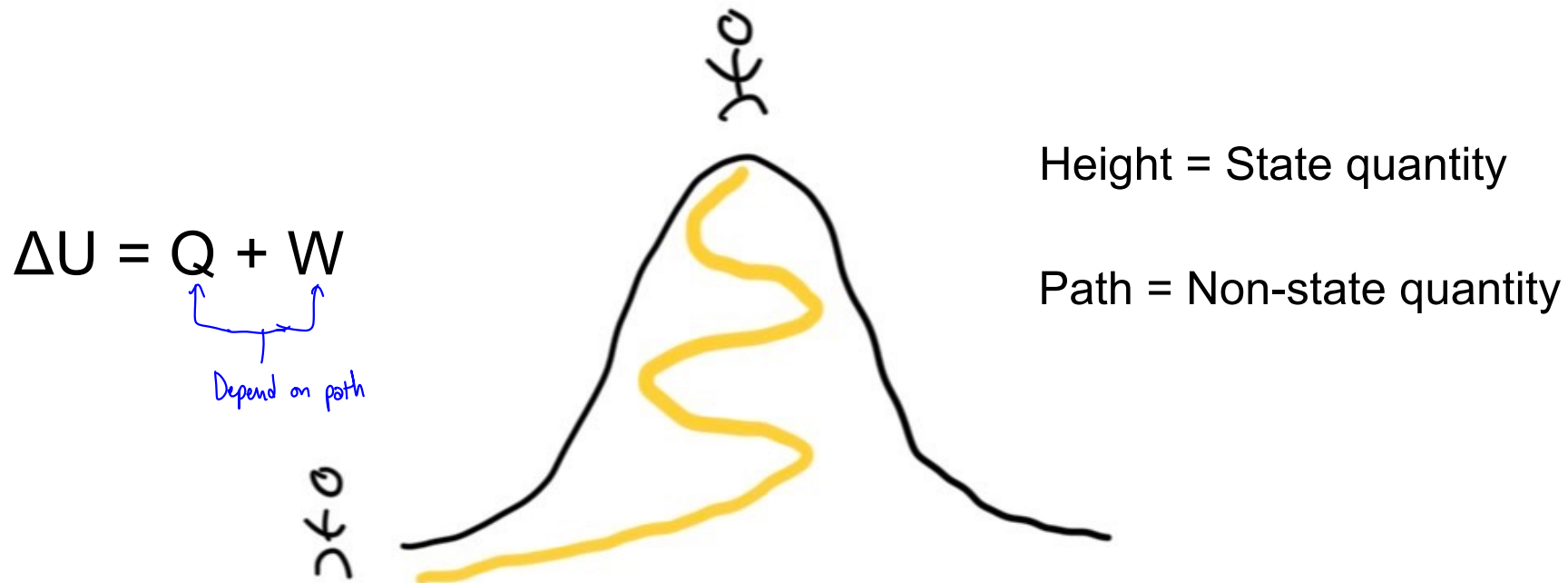
$$W_{\text{all}} = W_{\text{mechanical}} + \cancel{W_{\text{electric}}} + \cancel{W_{\text{surface}}}$$



State Function

U is state quantity

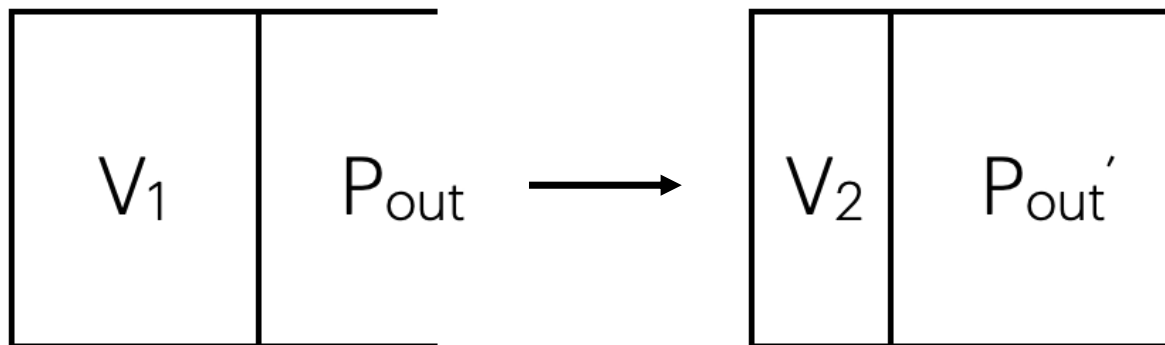
Q and W are non-state quantity



The first law of thermodynamics tells that U(state quantity) is always equal to the sum of non-state quantities.



Mechanical Work



$$W = - \int_{V_1}^{V_2} P_{out} dV$$

Why don't we use " P_{in} " instead of P_{out} ?

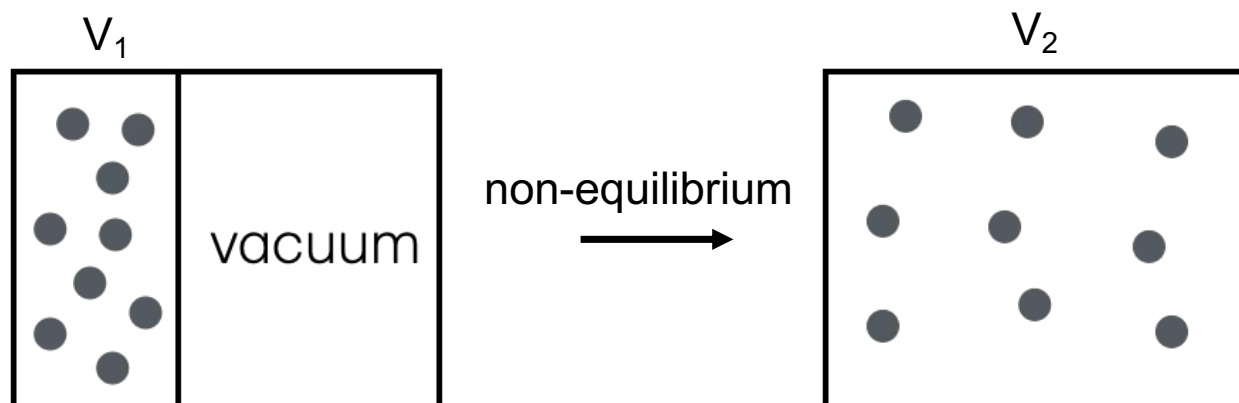
=> Inner pressure drastically changes. On the other hand, surrounding is enough huge and can be defined.

Handwritten notes in Thai:
 ค่าความดัน เปลี่ยนเสมอ = $\propto V$
 ค่าความดันที่



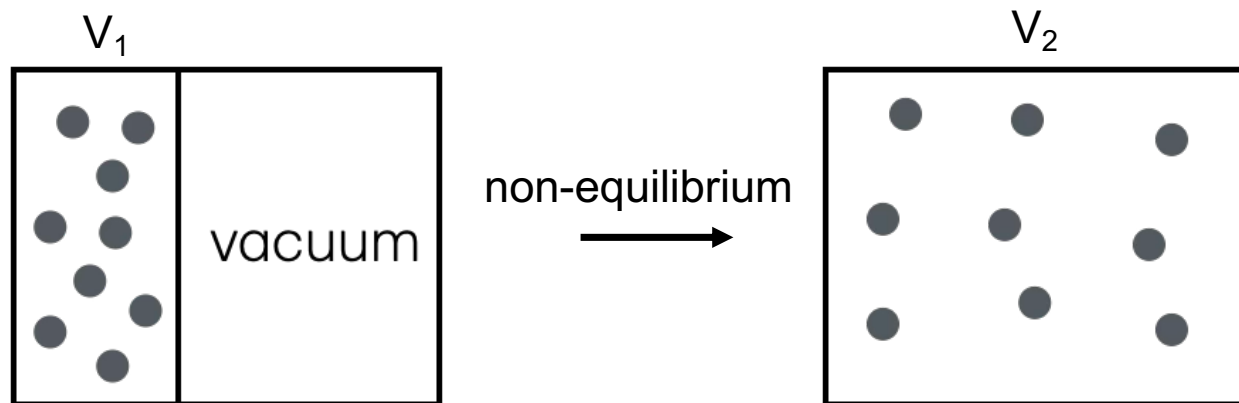
Mechanical Work

Ex) free expansion



Mechanical Work

Ex) free expansion

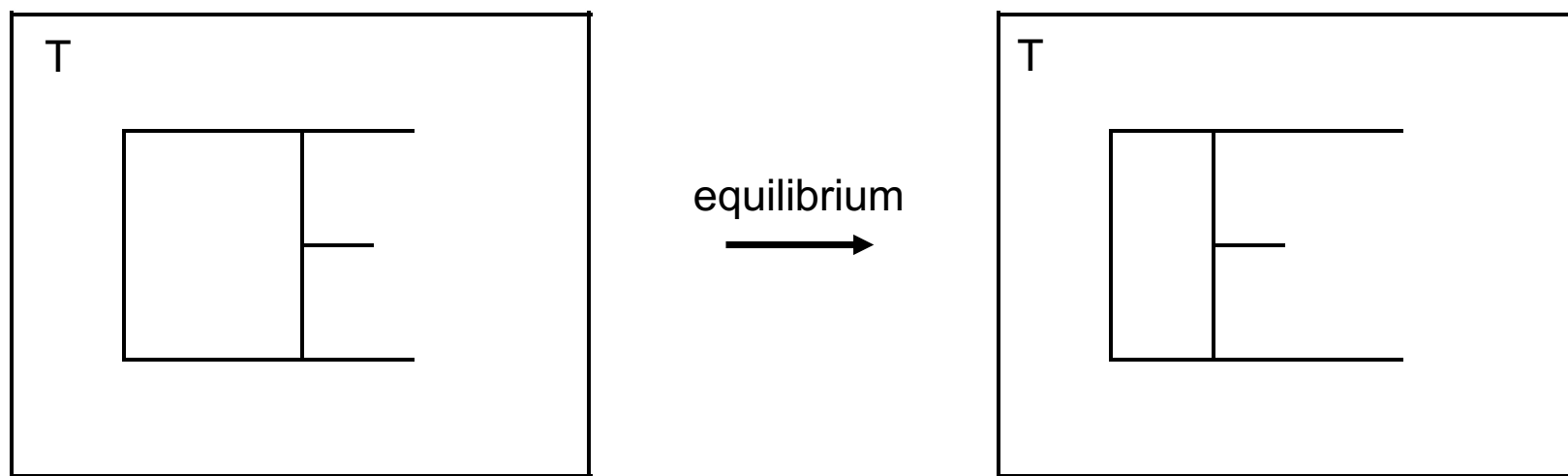


$$W = - \int_{V_1}^{V_2} P_{\text{out}} dV = 0$$



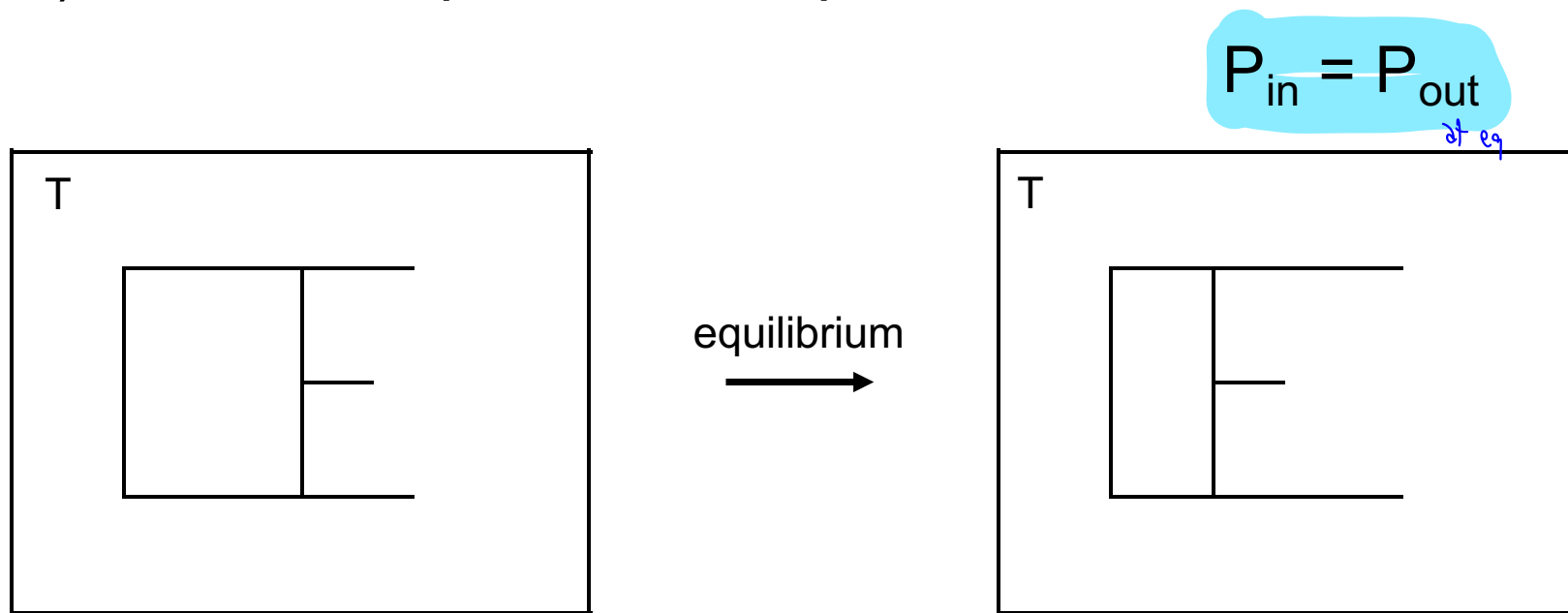
Mechanical Work

Ex) isothermal quasistatic expansion



Mechanical Work

Ex) isothermal quasistatic expansion



$$W = - \int_{V_1}^{V_2} P_{out} dV$$

$$P = nRT/V$$

$$= - \int_{V_1}^{V_2} P_{in} dV \quad \text{Ideal gas : } P_{in} = \frac{nRT}{V_{in}}$$

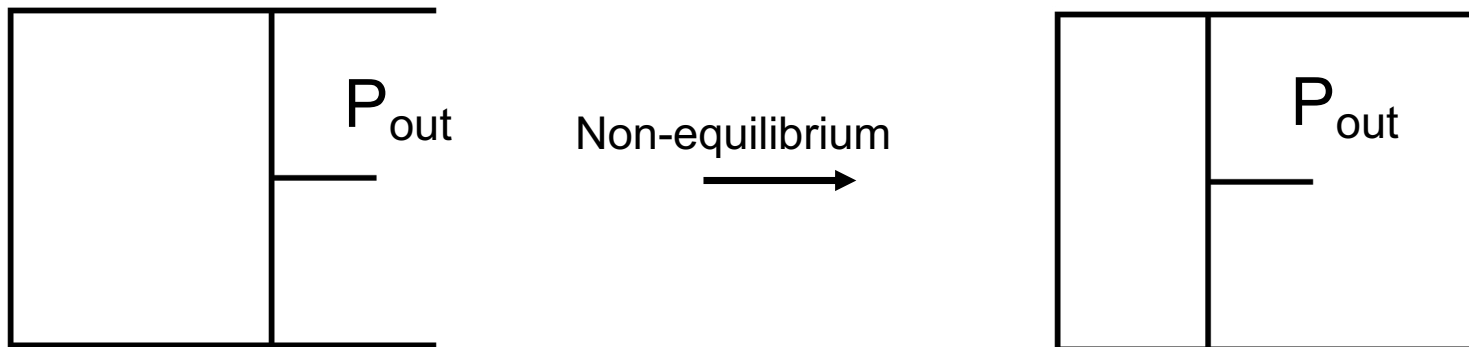
$$= -nRT \int_{V_1}^{V_2} 1/V dV = nRT \ln(V_2/V_1)$$

Thermochemistry



Mechanical Work

Ex) isobaric (constant pressure) expansion



$$W = - \int_{V_1}^{V_2} P_{out} dV$$
$$= -P_{out} \int_{V_1}^{V_2} dV = -P_{out} \cdot \Delta V$$

Enthalpy

$$\Delta U = Q + W$$



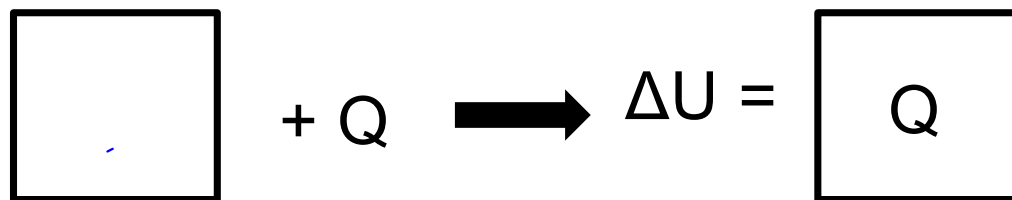
Thermochemistry

Enthalpy

Isovolumetric process ($V = \text{constant}$)

$$\Delta U = Q + W = Q$$

$$W = PdV = 0$$



=> In an isochoric (isovolumetric) process, the energy entering and leaving the system is equal to the change in internal energy

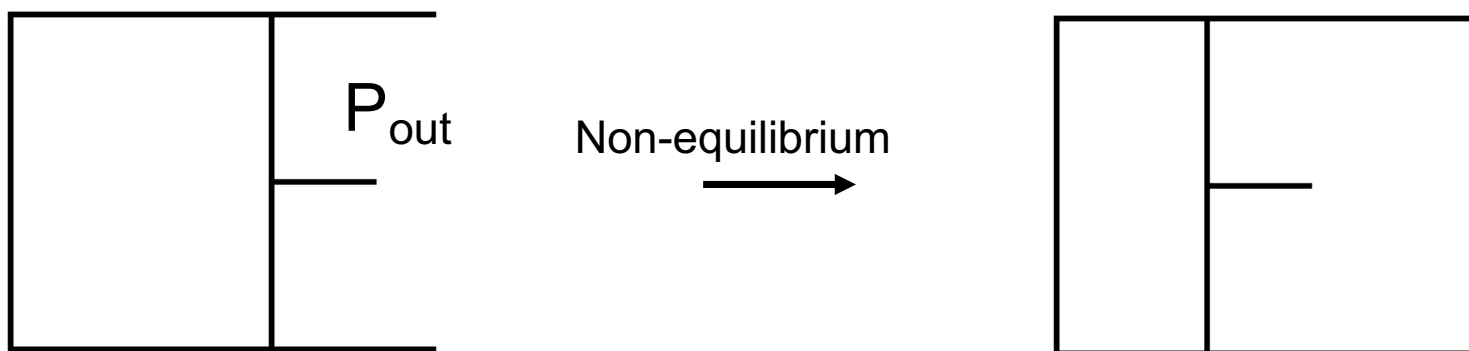
=> ^{molecules' energy} Internal energy cannot be directly measured.
Instead, ^{initial vs final state} heat can be measured.

Thermochemistry



Enthalpy

Isobaric process ($P = \text{constant}$)



$$H = U + PV$$

$$H' = U' + PV'$$

state func

$$\Delta H = H' - H$$

Enthalpy *Internal energy*

$$= (U' - U) + P(V' - V)$$

$$= \Delta U + P\Delta V$$

+W

$$= Q$$

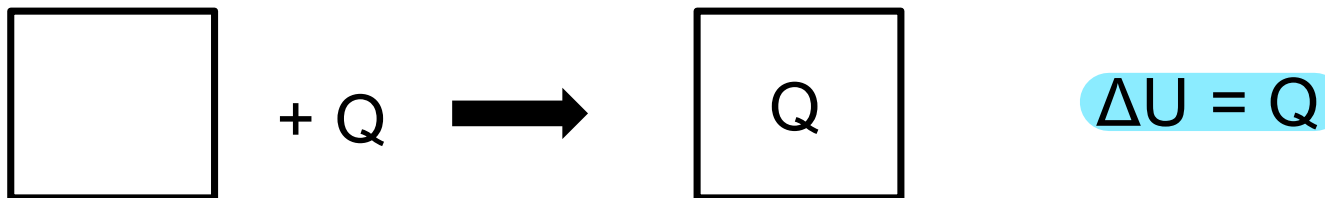
$$\Delta U = Q + W, P\Delta V = -W$$

Thermochemistry

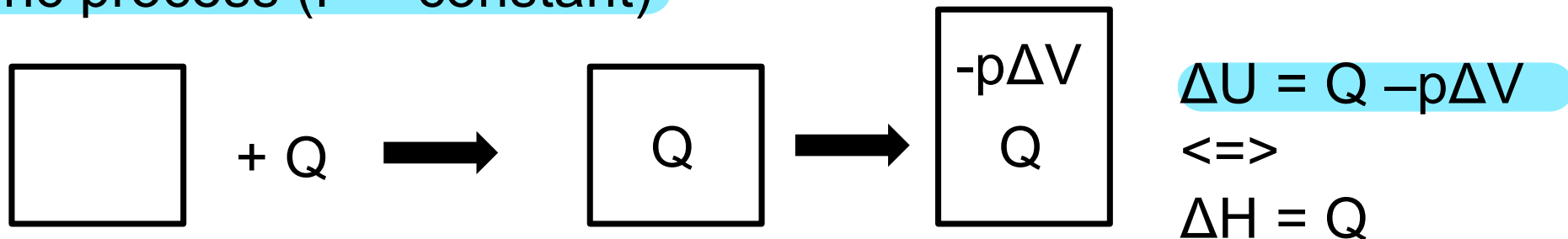


Enthalpy

Isovolumetric process ($V = \text{constant}$)



Isobaric process ($P = \text{constant}$)

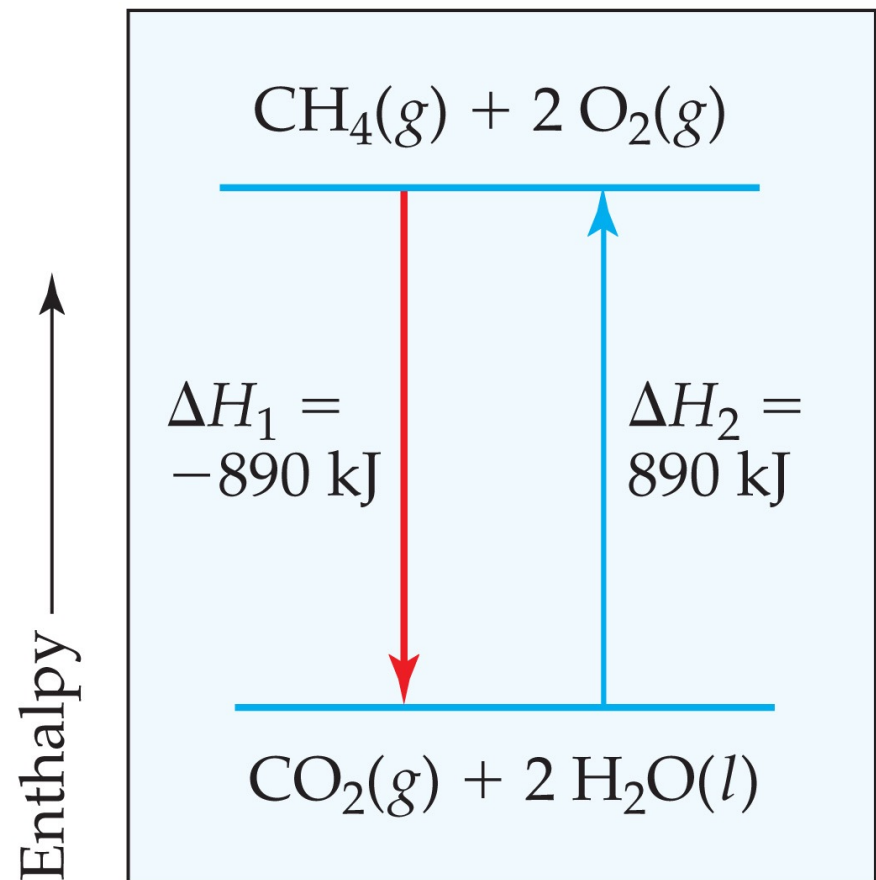


Thermochemistry

Enthalpy of Reaction

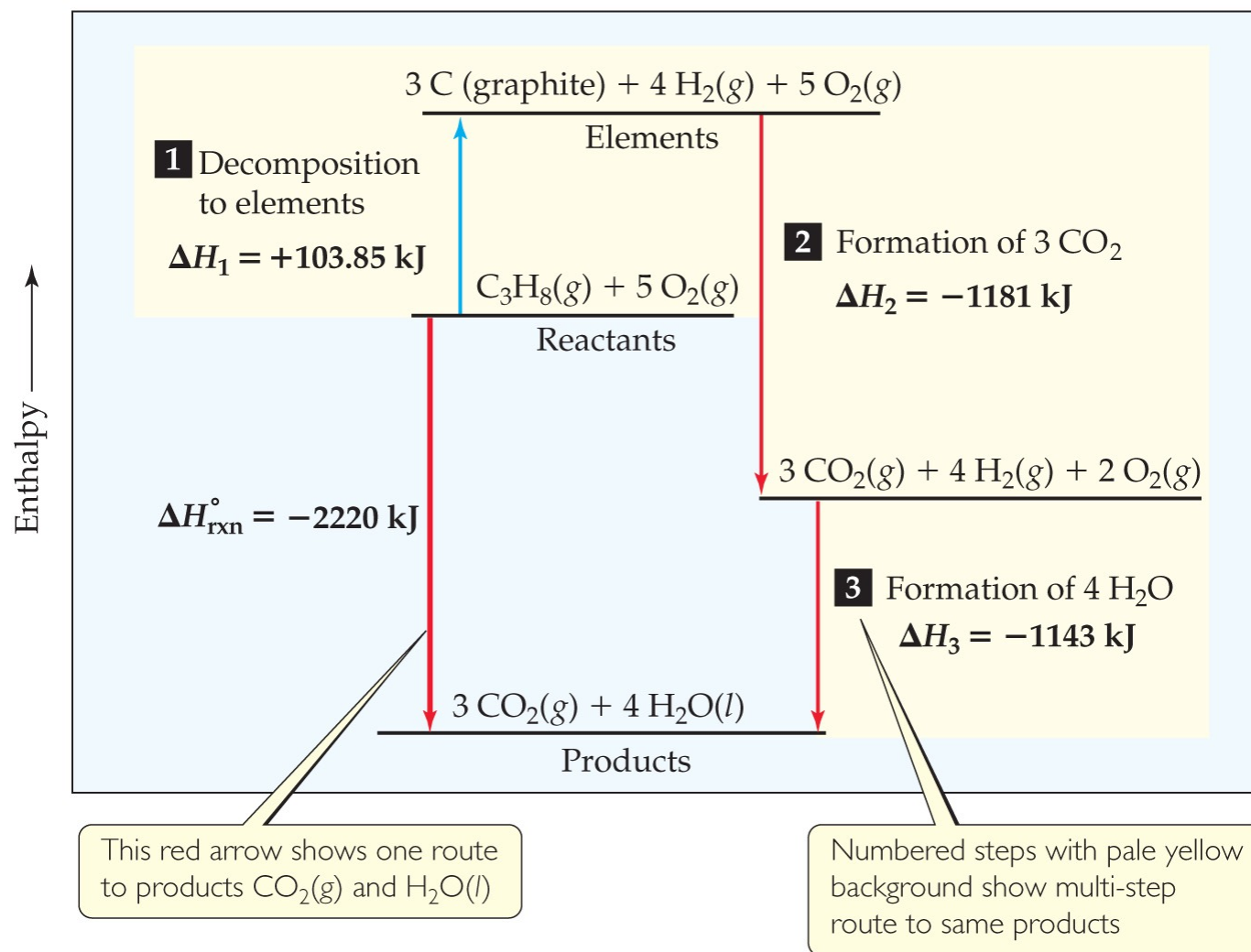
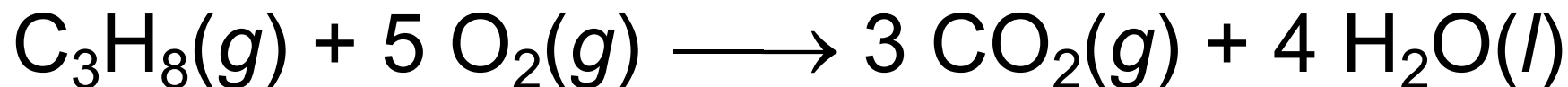
The *change* in enthalpy, ΔH , is the enthalpy of the products minus the enthalpy of the reactants:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$



Thermochemistry

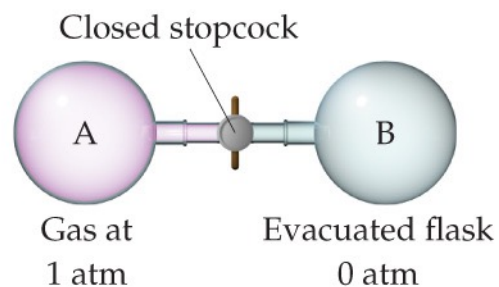
Calculation of ΔH



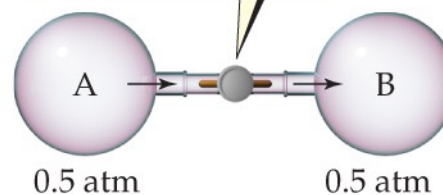
Thermochemistry



Spontaneous Processes

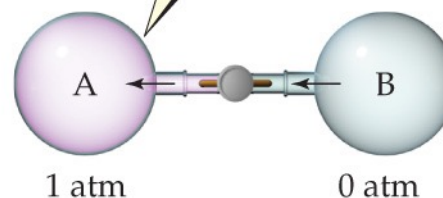


When stopcock opens, gas expands to occupy both flasks



This process is spontaneous

All gas molecules move back into flask A



This process is not spontaneous

Thermochemistry



Entropy

- **Entropy** can be thought of as a measure of the randomness of a system.
- It is a state function:

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

- It can be found by heat transfer from surroundings at a given temperature:

$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{constant } T)$$



Second Law of Thermodynamics

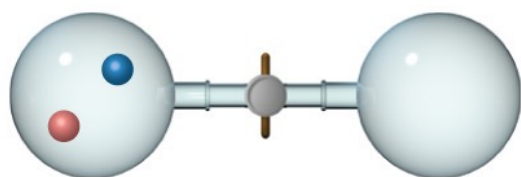
- The entropy of the universe increases in any spontaneous processes.
- This results in the following relationships:

Reversible Process: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$

Irreversible Process: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$

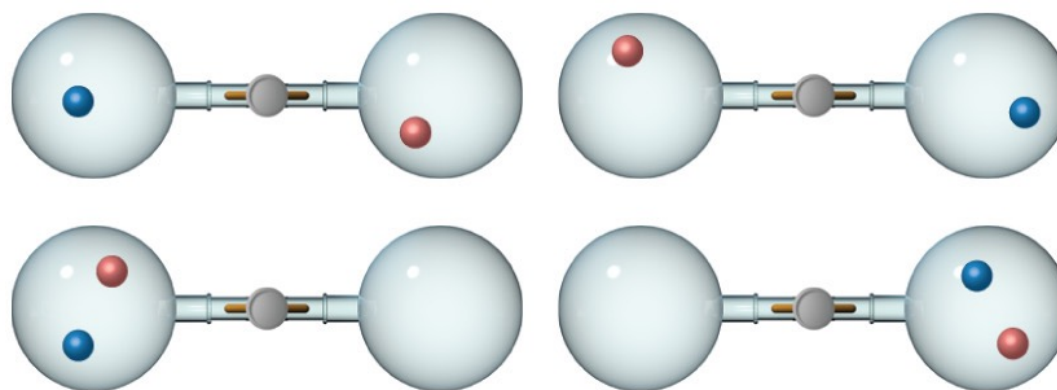


Entropy on the Molecular Scale



(a)

The two molecules are colored red and blue to keep track of them.



(b)

Four possible arrangements (microstates) once the stopcock is opened.

$$(1/2)^2$$

$$(1/2)^{6.02 \times 10^{23}}$$



Boltzmann's Use of Microstates

- Because there are so many possible microstates, we can't look at every picture.
- W represents the number of microstates.
- Entropy is a measure of how many microstates are associated with a particular macroscopic state.
- The connection between the number of microstates and the entropy of the system is:

$$S = k \ln W$$



Meaning of Entropy

$$S = k \ln W$$

(W = number of microstates)
=> randomness or disorder



Meaning of Entropy

$$S = k \ln W$$

(W = number of microstates)
=> randomness or disorder

It is heat that causes the chaotic movements!!



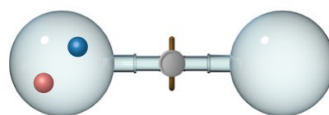
Meaning of Entropy

$$S = k \ln W$$

(W = number of microstates)
=> randomness or disorder

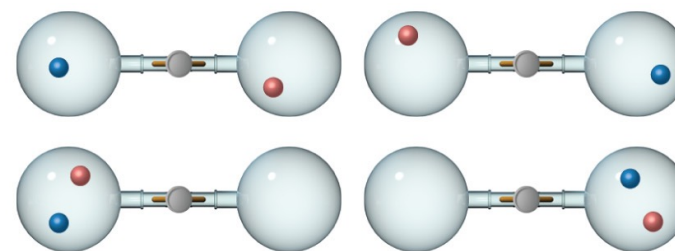
It is heat that causes the chaotic movements!!

$$\Delta E = q + W$$



(a)

The two molecules are colored red and blue to keep track of them.



(b)

Four possible arrangements (microstates) once the stopcock is opened.

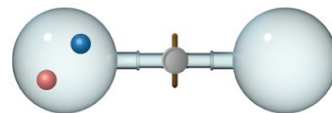
Meaning of Entropy

$$S = k \ln W$$

(W = number of microstates)
=> randomness or disorder

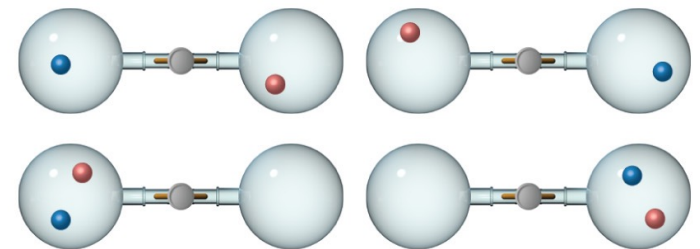
It is heat that causes the chaotic movements!!

$$\Delta E = q + W$$



(a)

The two molecules are colored red and blue to keep track of them.



(b)

Four possible arrangements (microstates) once the stopcock is opened.

$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{constant } T)$$



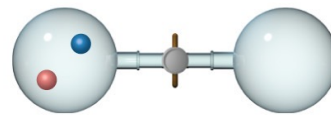
Meaning of Entropy

$$S = k \ln W$$

(W = number of microstates)
=> randomness or disorder

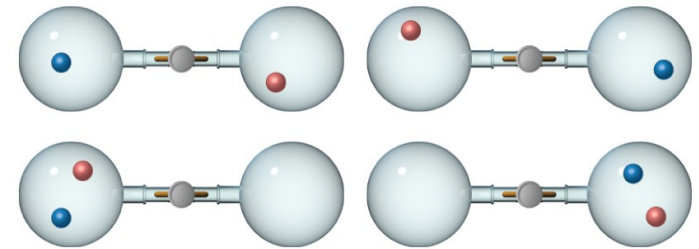
It is heat that causes the chaotic movements!!

$$\Delta E = q + W$$



(a)

The two molecules are colored red and blue to keep track of them.



(b)

Four possible arrangements (microstates) once the stopcock is opened.

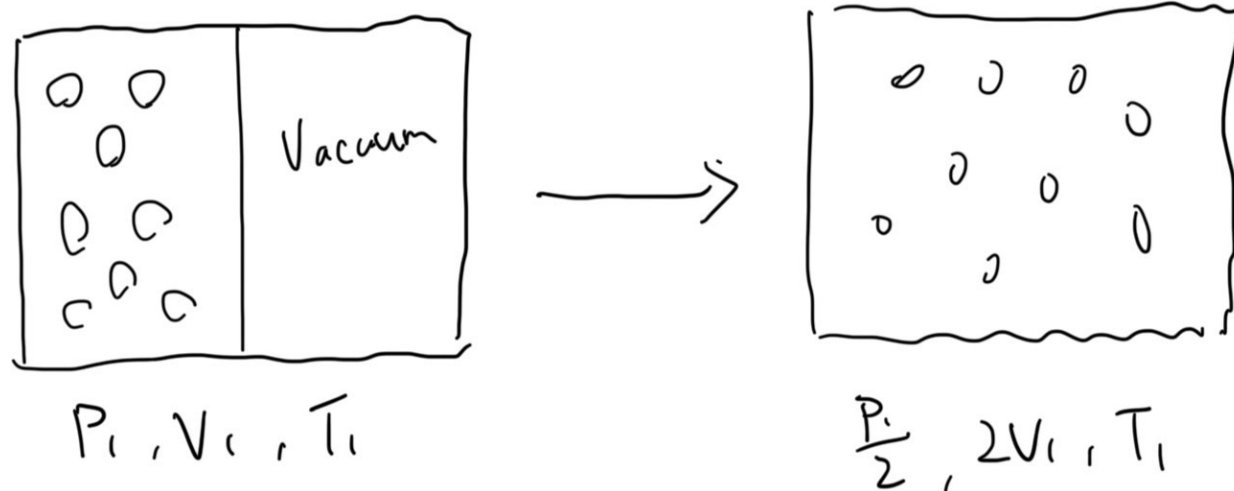
$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{constant } T)$$

Original disorder



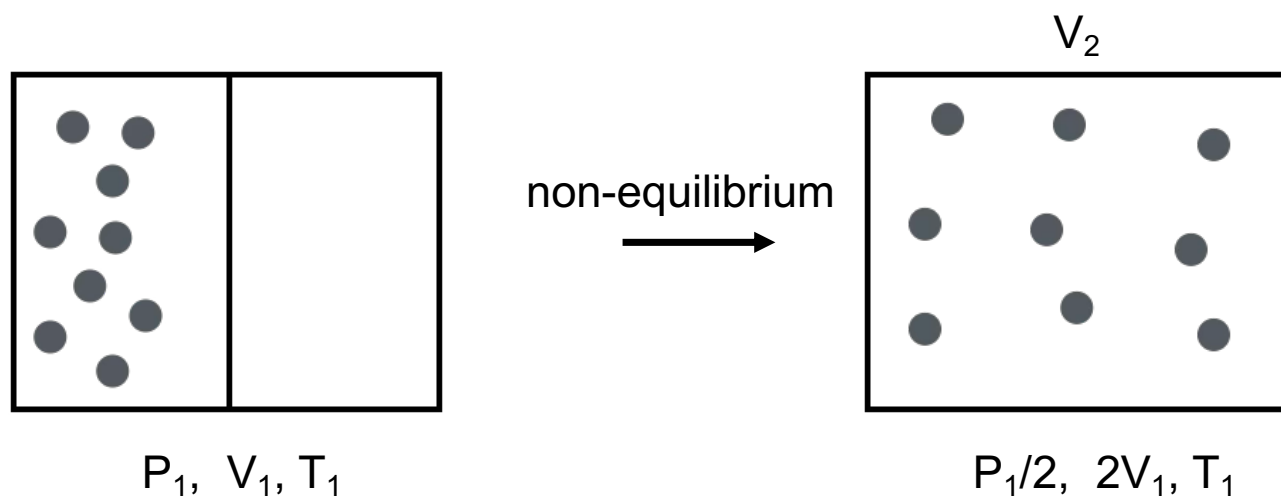
Calculation of Entropy

Adiabatic free expansion



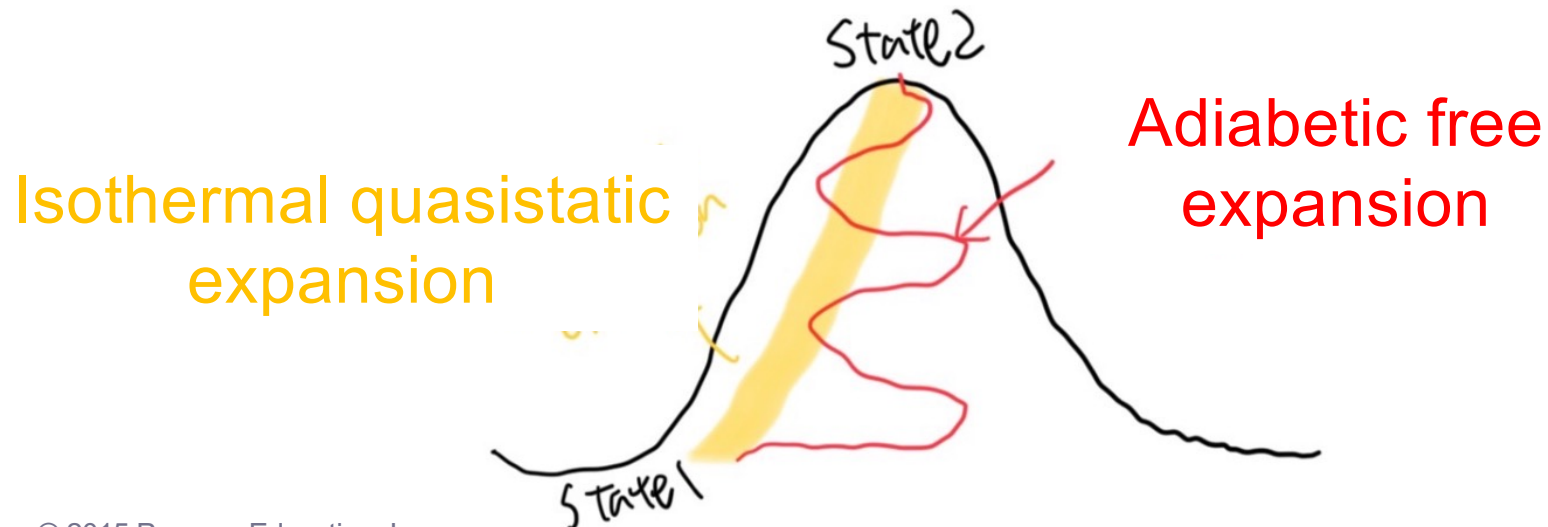
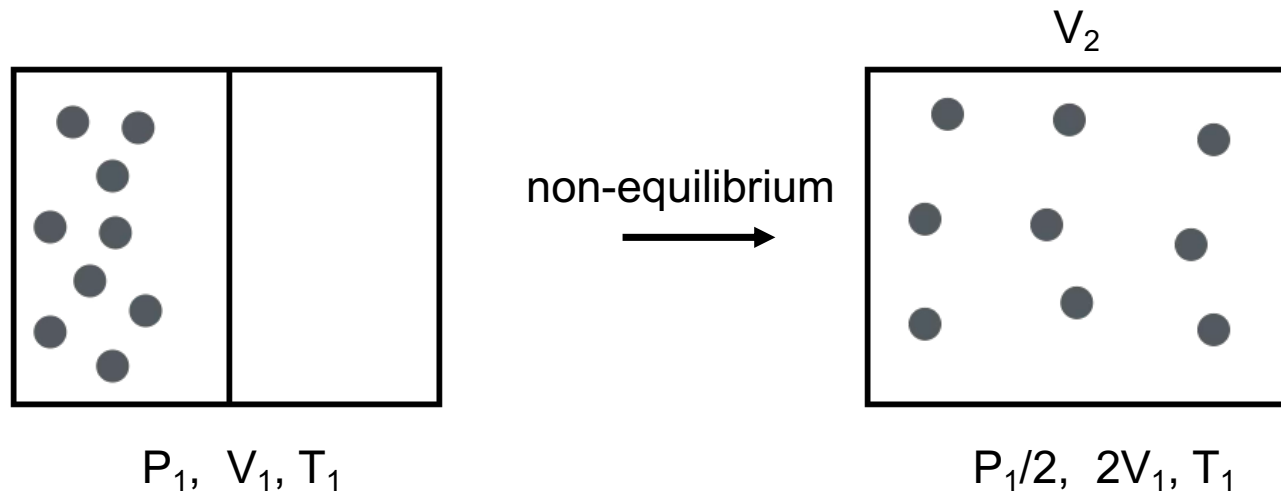
Calculation of Entropy

Adiabatic free expansion



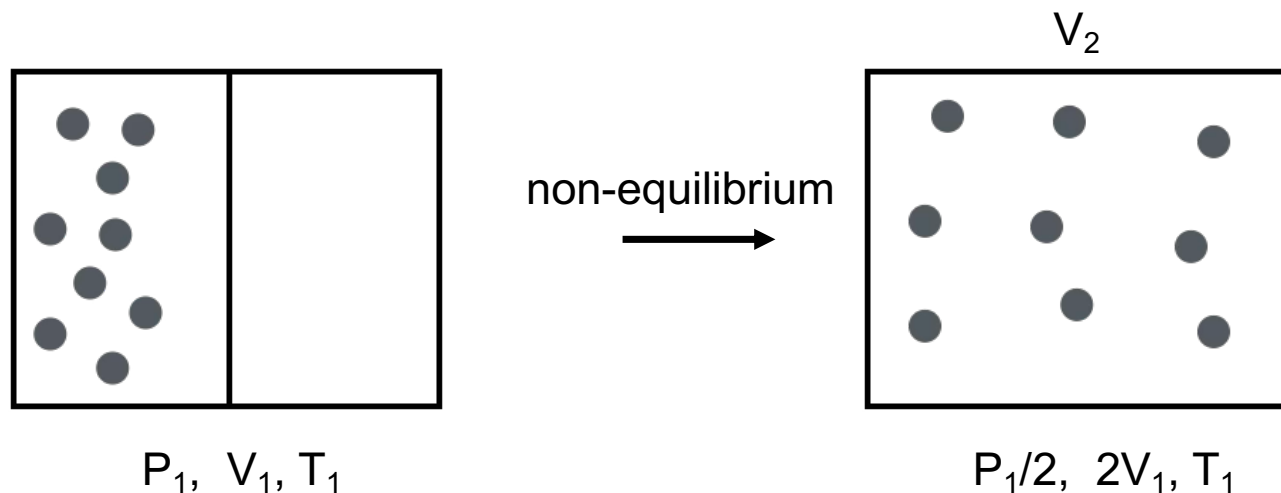
Calculation of Entropy

Adiabatic free expansion



Calculation of Entropy

Adiabatic free expansion



$$W = - \int_{V_1}^{2V_1} P_{\text{out}} dV$$

$$= - \int_{V_1}^{2V_1} P_{\text{in}} dV$$

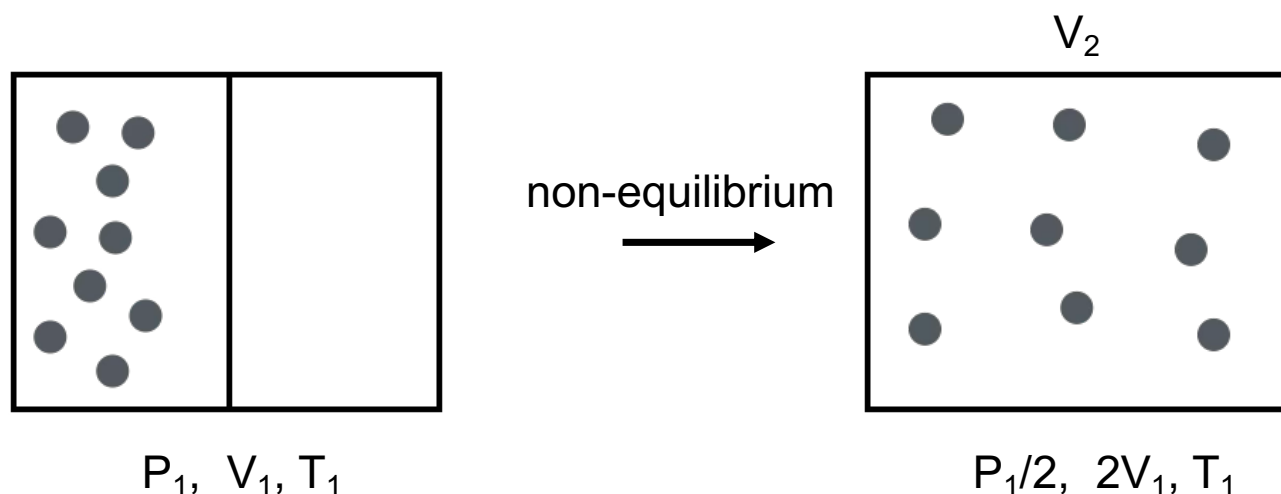
$$= -nRT \int_{V_1}^{2V_1} \frac{1}{V} dV = nRT_1 \ln 2$$

$$P = nRT/V$$



Calculation of Entropy

Adiabatic free expansion



$$\begin{aligned}
 W &= - \int_{V_1}^{2V_1} P_{\text{out}} dV \\
 &= - \int_{V_1}^{2V_1} P_{\text{in}} dV \\
 &= -nRT \int_{V_1}^{2V_1} \frac{1}{V} dV = nRT_1 \ln 2
 \end{aligned}$$

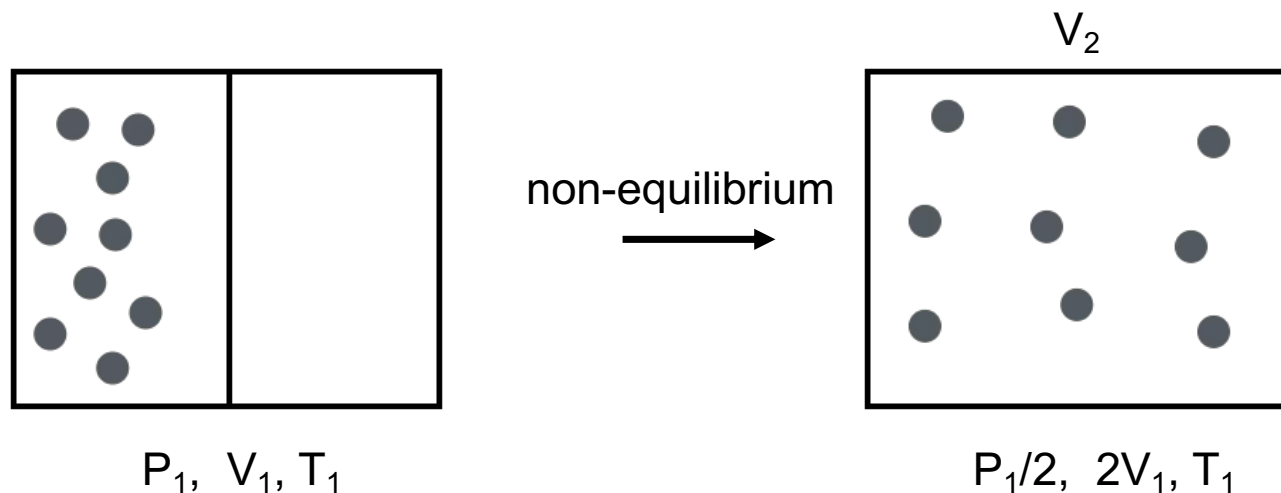
$$\begin{aligned}
 Q &= \Delta U - W \quad (\Delta U = 0) \\
 &= nRT_1 \ln 2
 \end{aligned}$$

$$\Delta S = \int dQ/T = Q/T_1 = nRT_1 \ln 2$$

Thermochemistry

Calculation of Entropy

Adiabatic free expansion



$$\Delta S_{\text{uni}} = \Delta S_{\text{sys}} + \Delta S_{\text{sur}}$$

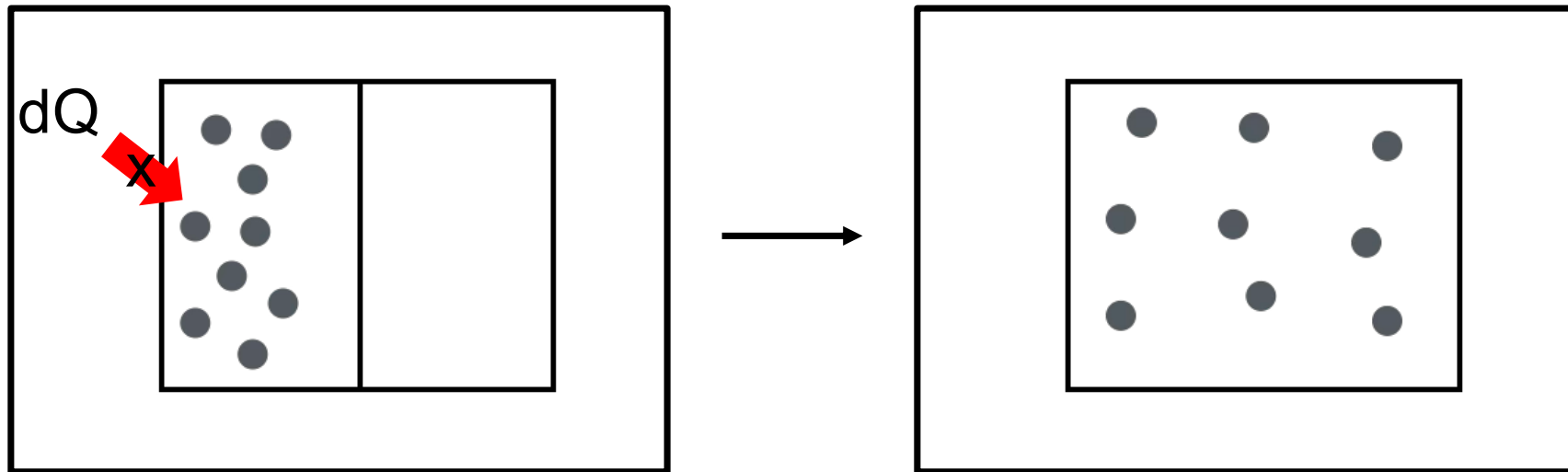
$$\Delta S_{\text{sur}} = \int -dQ/T = 0$$

$$\Delta S_{\text{uni}} = \Delta S_{\text{sys}} > 0$$



Calculation of Entropy

Adiabatic free expansion



$$\Delta S_{\text{sys}} = nRT \ln 2$$

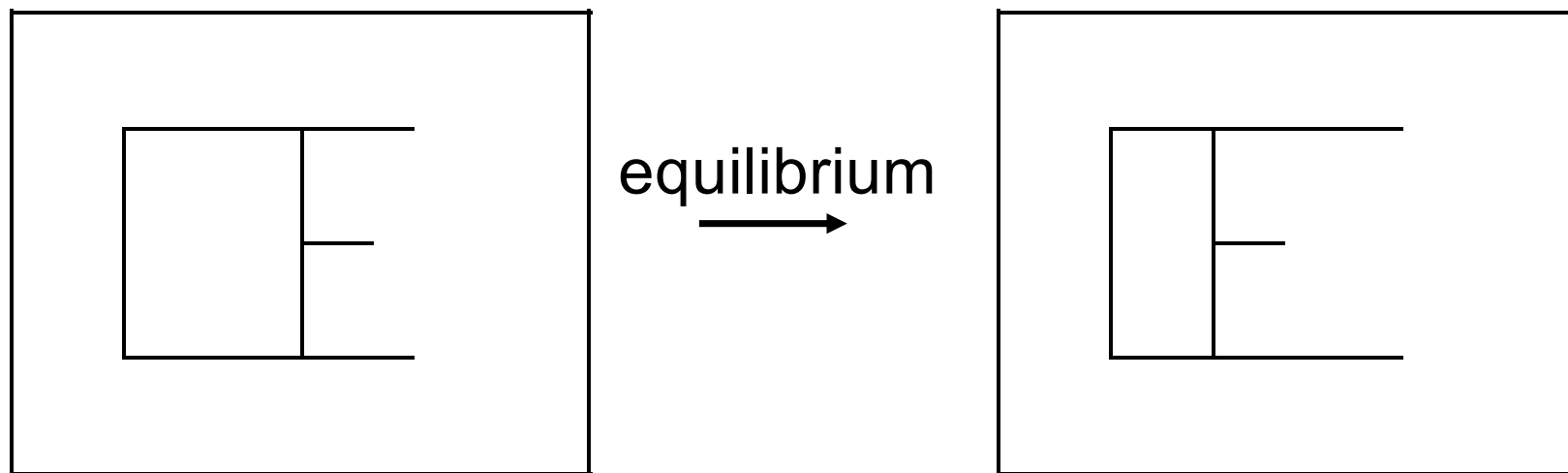
$$\Delta S_{\text{sur}} = 0$$

$$\Delta S_{\text{uni}} = \Delta S_{\text{sys}} > 0$$



Calculation of Entropy

Isothermal quasistatic expansion



$$\Delta s_{\text{sys}} = nRT \ln 2$$

$$\Delta S_{\text{sur}} = - nRT \ln 2$$

$$\Delta S_{\text{uni}} = 0$$



Entropy Changes in Surroundings

- Heat that flows into or out of the system changes the entropy of the surroundings.
- For an isothermal process

$$\Delta S_{\text{surr}} = \frac{-q_{\text{sys}}}{T}$$

- At constant pressure, q_{sys} is simply ΔH° for the system.

$$\Delta S_{\text{surr}} = \frac{-\Delta H_{\text{sys}}}{T} \quad [\text{at constant P}]$$



Entropy Change in the Universe

- The universe is composed of the system and the surroundings.
- Therefore,

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

- For spontaneous processes

$$\Delta S_{\text{universe}} > 0$$

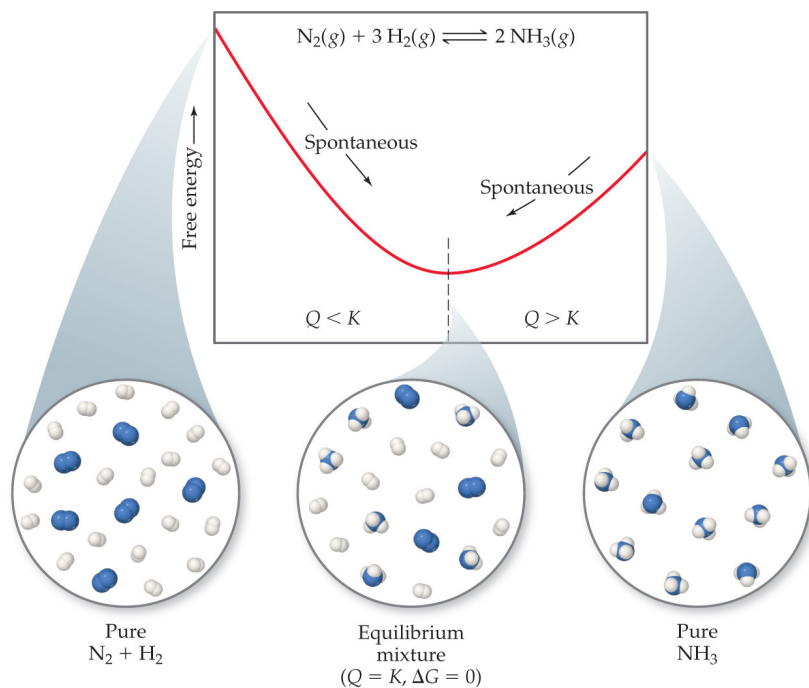


Total Entropy and Spontaneity

- $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$
- Substitute for the entropy of the surroundings:
 - $\Delta S_{\text{universe}} = \Delta S_{\text{system}} - \Delta H_{\text{system}}/T$
- Multiply by $-T$:
 - $-T\Delta S_{\text{universe}} = -T\Delta S_{\text{system}} + \Delta H_{\text{system}}$
- Rearrange:
 - $-T\Delta S_{\text{universe}} = \Delta H_{\text{system}} - T\Delta S_{\text{system}}$
- Call $-T\Delta S_{\text{universe}}$ the **Gibbs Free Energy (ΔG)**:
 - **$\Delta G = \Delta H - T\Delta S$**



Gibbs Free Energy

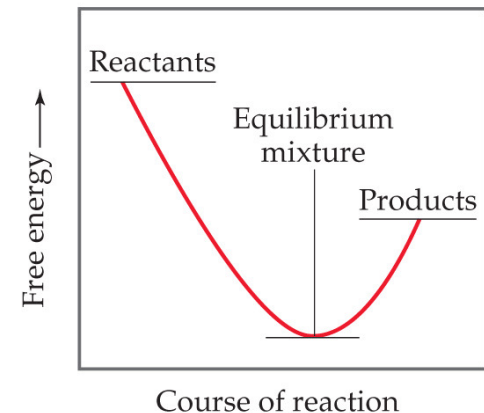
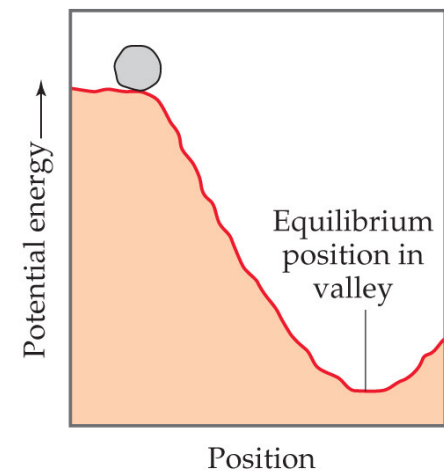


1. If ΔG is negative, the forward reaction is spontaneous.
2. If ΔG is 0, the system is at equilibrium.
3. If ΔG is positive, the reaction is spontaneous in the reverse direction.

Standard Free Energy Changes

Analogous to standard enthalpies of formation are standard free energies of formation, ΔG_f° :

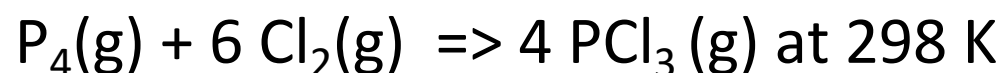
where n and m are the stoichiometric coefficients.



Thermochemistry

Standard Free Energy Changes

Calculate the standard free energy for the following reaction



$$\begin{aligned}\Delta G_{\text{rxn}}^{\circ} &= 4 \Delta G_f^{\circ}[\text{PCl}_3(\text{g})] - \Delta G_f^{\circ}[\text{P}_4(\text{g})] - 6 \Delta G_f^{\circ}[\text{Cl}_2(\text{g})] \\ &= (4 \text{ mol})(-269.6 \text{ kJ/mol}) - (1 \text{ mol})(24.4 \text{ kJ/mol}) - 0 \\ &= -1102.8 \text{ kJ}\end{aligned}$$



Free Energy Changes

Table 19.3 How Signs of ΔH and ΔS Affect Reaction Spontaneity

ΔH	ΔS	$-T\Delta S$	$\Delta G = \Delta H - T\Delta S$	Reaction Characteristics	Example
–	+	–	–	Spontaneous at all temperatures	$2 \text{O}_3(g) \longrightarrow 3 \text{O}_2(g)$
+	–	+	+	Nonspontaneous at all temperatures	$3 \text{O}_2(g) \longrightarrow 2 \text{O}_3(g)$
–	–	+	+ or –	Spontaneous at low T ; nonspontaneous at high T	$\text{H}_2\text{O}(l) \longrightarrow \text{H}_2\text{O}(s)$
+	+	–	+ or –	Spontaneous at high T ; nonspontaneous at low T	$\text{H}_2\text{O}(s) \longrightarrow \text{H}_2\text{O}(l)$

- How does ΔG change with temperature?
- $\Delta G = \Delta H - T\Delta S$
- Since reactions are spontaneous if $\Delta G < 0$, the sign of enthalpy and entropy and the magnitude of the temperature matters to spontaneity.



Free Energy and Equilibrium

Under any conditions, standard or nonstandard, the free energy change can be found this way:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Here, Q is reaction quotient.



$$Q_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$



Free Energy and Equilibrium

Under any conditions, standard or nonstandard, the free energy change can be found this way:

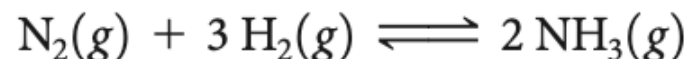
$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

(Under standard conditions, concentrations are 1 *M*, so $Q = 1$ and $\ln Q = 0$; the last term drops out.)



Free Energy and Equilibrium

Calculate ΔG at 298 K for a mixture of 1.0 atm N_2 , 3.0 atm H_2 , and 0.50 atm NH_3 being used in the Haber process:



$$Q = \frac{P_{\text{NH}_3}^2}{P_{\text{N}_2} P_{\text{H}_2}^3} = \frac{(0.50)^2}{(1.0)(3.0)^3} = 9.3 \times 10^{-3}$$

$$\begin{aligned}\Delta G &= \Delta G^\circ + RT \ln Q \\ &= (-33.3 \text{ kJ/mol}) + (8.314 \text{ J/mol-K})(298 \text{ K})(1 \text{ kJ}/1000 \text{ J}) \ln(9.3 \times 10^{-3}) \\ &= (-33.3 \text{ kJ/mol}) + (-11.6 \text{ kJ/mol}) = -44.9 \text{ kJ/mol}\end{aligned}$$

Thermochemistry



Free Energy and Equilibrium

- At equilibrium, $Q = K$, and $\Delta G = 0$.
- The equation becomes

$$0 = \Delta G^\circ + RT \ln K$$

- Rearranging, this becomes

$$\Delta G^\circ = -RT \ln K$$

or

$$K = e^{-\Delta G^\circ / RT}$$



Free Energy and Equilibrium



Use this value of ΔG° to calculate the equilibrium constant for the process at 25 °C.

$$K = e^{-\Delta G^\circ/RT} = e^{-(-33,300 \text{ J/mol})/(8.314 \text{ J/mol-K})(298 \text{ K})} = e^{13.4} = 7 \times 10^5$$

